

GIELAS, Sweden

WMO: 021100

Lat: 65.327N

Lon: 15.065E

Elev: 577

StdP: 94.58

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.9	-30.7	-37.4	0.1	-33.9	-33.7	0.2	-30.7	7.9	0.2	7.0	-3.2	1.0	250	0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	22.7	15.3	20.7	14.4	18.6	13.6	16.6	21.1	15.3	19.3	14.1	17.8	2.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.8	11.3	18.5	13.5	10.4	16.9	12.4	9.6	15.9	48.6	21.5	44.9	19.3	41.4	17.8	20.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.4	5.4	4.7	-37.9	24.4	2.4	3.1	-39.6	26.6	-41.0	28.4	-42.4	30.2	-44.1	32.4		
			-38.4	17.7	2.3	1.6	-40.1	18.8	-41.4	19.8	-42.7	20.7	-44.4	21.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.5	-11.6	-11.2	-7.7	-1.6	3.7	8.7	11.8	9.9	5.9	0.2	-6.0	-8.5
	DBStd	9.84	8.45	8.18	6.62	4.38	3.39	3.34	3.13	2.92	3.18	4.22	6.80	7.88
	HDD10.0	3968	669	594	548	349	199	65	17	38	127	304	481	575
	HDD18.3	6873	928	828	807	599	454	290	204	262	374	563	731	833
	CDD10.0	137	0	0	0	0	3	25	72	34	3	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	38	0	0	0	0	0	8	28	2	0	0	0	0
	CDH26.7	3	0	0	0	0	0	0	3	0	0	0	0	0
Wind	WSAvg	2.0	2.0	2.0	2.3	2.2	2.2	2.2	1.9	1.7	2.0	2.1	1.8	2.1
Precipitation	PrecAvg	693	63	56	48	32	37	63	93	77	56	54	52	61
	PrecMax	945	121	164	115	58	82	127	175	168	149	87	111	105
	PrecMin	540	10	8	8	6	13	10	36	15	17	24	15	23
	PrecStd	129	29	38	26	14	18	29	40	42	28	18	26	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.8	4.4	5.8	12.6	20.6	24.6	26.3	23.4	17.8	11.0	7.5	4.4
		MCWB	2.6	2.8	3.0	6.1	11.8	15.3	17.0	16.1	11.5	8.5	6.7	3.2
	2%	DB	2.0	2.7	3.7	8.8	16.8	21.5	23.4	21.1	14.8	8.8	4.9	3.1
		MCWB	1.0	1.6	1.8	4.3	10.1	13.1	15.7	15.4	10.9	6.8	3.8	2.2
	5%	DB	0.7	1.1	2.6	6.5	13.7	18.7	21.3	18.8	12.8	7.3	3.0	2.0
		MCWB	-0.1	0.3	1.1	3.4	8.7	12.2	14.9	14.4	10.0	5.8	2.0	1.2
	10%	DB	-1.2	-0.5	1.5	4.8	10.3	16.2	19.2	16.5	11.4	6.1	1.6	0.7
		MCWB	-1.9	-1.3	0.1	2.4	6.7	11.1	14.4	13.0	9.3	4.9	0.8	-0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	2.9	3.4	6.7	12.5	16.9	18.4	17.9	13.1	9.1	6.4	3.3
		MCDB	3.8	4.1	4.9	11.1	19.1	22.6	23.0	21.2	16.0	10.6	7.0	4.0
	2%	WB	1.2	1.8	2.1	5.1	10.7	14.6	17.0	16.1	11.6	7.2	3.5	2.3
		MCDB	1.8	2.7	3.4	8.0	15.5	19.6	22.3	20.0	13.9	8.5	4.5	3.0
	5%	WB	0.0	0.4	1.2	3.7	8.9	13.0	15.8	14.8	10.5	6.2	1.9	1.2
		MCDB	0.7	1.2	2.5	5.9	13.6	17.5	19.8	18.2	12.4	7.3	2.7	1.9
	10%	WB	-1.8	-1.4	0.1	2.5	7.0	11.5	14.7	13.5	9.5	5.1	0.8	-0.1
		MCDB	-1.2	-0.6	1.5	4.6	10.5	15.3	18.6	16.4	11.2	6.1	1.5	0.6
Mean Daily Temperature Range	5% DB	MDBR	9.2	10.4	11.7	9.9	9.3	10.5	11.0	10.5	8.6	6.3	7.0	8.4
		MCDBR	6.8	6.8	9.4	12.5	16.3	16.3	15.8	15.3	11.6	7.5	5.7	7.2
		MCWBR	6.5	5.9	7.8	8.8	10.5	9.4	8.8	10.1	7.9	5.8	5.2	6.7
	5% WB	MCDBR	6.4	6.9	8.1	10.3	15.4	14.0	13.9	13.4	10.4	7.1	5.7	7.3
		MCWBR	6.2	6.2	6.9	7.5	10.5	9.3	8.9	9.4	7.3	5.6	5.2	6.7
Clear-Sky Solar Irradiance	taub	0.174	0.246	0.278	0.306	0.316	0.317	0.328	0.315	0.291	0.269	N/A	0.177	
	taud	1.929	2.181	2.230	2.260	2.417	2.518	2.504	2.526	2.578	2.412	N/A	1.941	
	Ebn at Noon	339	655	801	863	892	900	877	856	807	639	N/A	82	
	Edh at Noon	21	53	82	102	97	90	89	79	60	43	N/A	6	
All-Sky Solar Radiation	RadAvg	0.10	0.55	1.93	3.77	4.73	4.87	4.59	3.43	1.87	0.70	0.18	0.03	
	RadStd	0.01	0.12	0.23	0.31	0.35	0.44	0.56	0.36	0.23	0.08	0.02	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page